

Steganography Analysis and Hidden Data Detection using StegExpose

EXPERIMENT NO: 08

DATE:

1. AIM

To conduct a comprehensive steganalysis lifecycle on digital image evidence using the StegExpose command-line utility; to evaluate Least Significant Bit (LSB) statistical properties and detection algorithms across a target evidence directory; to compute quantitative suspicion scores based on configurable thresholds; and to automatically compile, format, and export the analytical findings into a structured CSV report for forensic documentation.

2. SOFTWARE / TOOLS REQUIRED

- **Operating System:** Windows, Linux, or macOS
- **Runtime Environment:** Java Runtime Environment (JRE)
- **Forensics Tool:** StegExpose (Java .jar executable)
- **Evidence Files:** Target image files (e.g., .png, .jpg, .bmp)

3. DESCRIPTION

StegExpose is a specialized steganalysis tool designed to detect the presence of hidden data within digital images. Unlike traditional steganography tools used to embed data, StegExpose serves a purely investigative role. It evaluates the statistical properties of an image to identify mathematical anomalies indicative of data hiding techniques, particularly Least Significant Bit (LSB) steganography.

The tool processes images and outputs a quantitative "suspect score" ranging from 0 to 1. By comparing this score against established thresholds, forensic investigators can mathematically estimate the likelihood that an image contains clandestine data, allowing them to triage suspicious files rapidly without needing the original extraction password.

4. PROCEDURE

Step 1: Open Command Prompt as Administrator

- Press the Windows Key, type cmd, right-click on **Command Prompt**, and select **Run as administrator**.

Step 2: Navigate to the Working Directory

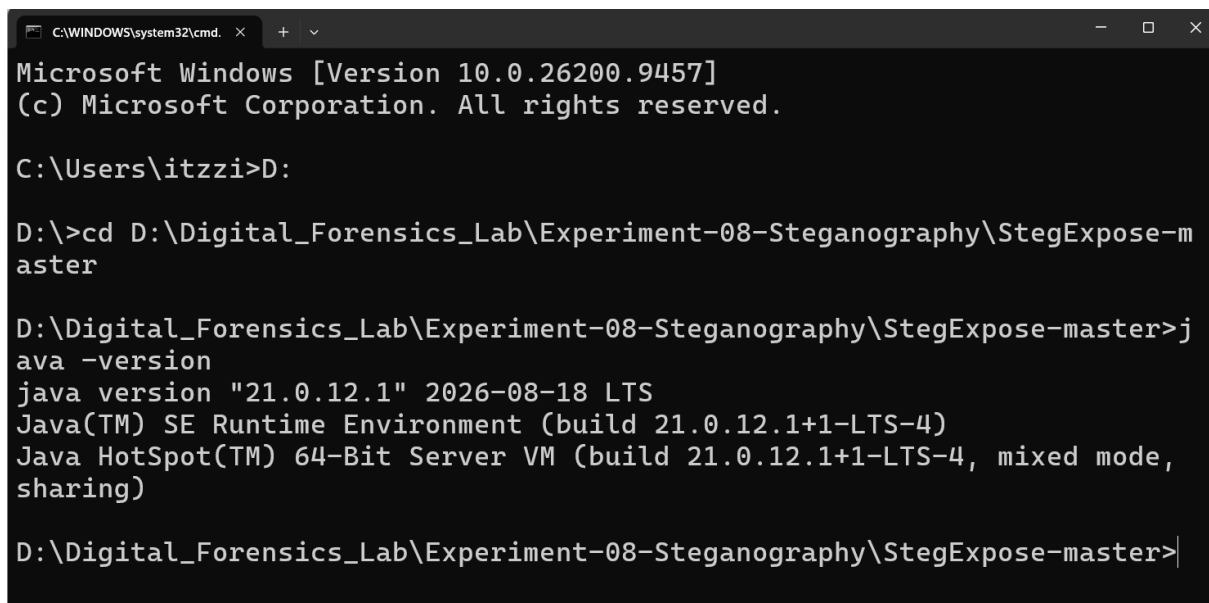
- Use the change directory (cd) command to point the terminal to your project folder where StegExpose.jar and your evidence directory are located:
- And check if java is available.

DOS

```
cd /d "D:\Digital_Forensics_Lab\Experiment-08-Steganography\StegExpose-master"
```

```
java -version
```

(it shows jre installed or not)



```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\itzzi>D:

D:\>cd D:\Digital_Forensics_Lab\Experiment-08-Steganography\StegExpose-master

D:\Digital_Forensics_Lab\Experiment-08-Steganography\StegExpose-master>java -version
java version "21.0.12.1" 2026-08-18 LTS
Java(TM) SE Runtime Environment (build 21.0.12.1+1-LTS-4)
Java HotSpot(TM) 64-Bit Server VM (build 21.0.12.1+1-LTS-4, mixed mode, sharing)

D:\Digital_Forensics_Lab\Experiment-08-Steganography\StegExpose-master>
```

Figure 1: Navigating to the working directory in the Command Prompt

Step 3: Execute StegExpose Analysis with CSV Output

- Run the StegExpose command, passing the path to your evidence folder, the execution speed parameter (default), the statistical threshold (0.2), and the desired output report filename (report.csv):

DOS

```
java -jar StegExpose.jar "D:\Digital_Forensics_Lab\Experiment-08-Steganography\Evidence"
default 0.2 report.csv
```

```
C:\WINDOWS\system32\cmd. x + v

D:\Digital_Forensics_Lab\Experiment-08-Steganography>cd /d "D:\Digital_Forensics_Lab\Experiment-08-Steganography\StegExpose-master"

D:\Digital_Forensics_Lab\Experiment-08-Steganography\StegExpose-master>j
ava -jar StegExpose.jar "D:\Digital_Forensics_Lab\Experiment-08-Steganog
raphy\Evidence" default 0.2 report.csv

D:\Digital_Forensics_Lab\Experiment-08-Steganography\StegExpose-master>|
```

Figure 2: Executing StegExpose via CLI and redirecting output to report.csv

Step 4: Verify and Review the Generated Report

- Once the execution finishes, navigate to your folder or open the file in Microsoft Excel/Notepad to verify the generated report.csv contents.
- Otherwise, directly open the csv file on excel.

DOS

notepad report.csv

	B	C	D	E	F	G	H
1							
2	Above stego thres	Secret message s	Primary Sets	Chi Square	Sample Pairs	RS analysis	Fusion (mean)
3	FALSE	26	1.54E-04	1.72E-05	0.002401273	0.002734389	0.001326793
4							
5							
6							
7							

Figure 3: Opening and reviewing the generated CSV report detailing LSB scores and classifications

5. OBSERVATIONS

- **Batch Processing:** StegExpose scanned all image files within the target evidence folder sequentially.
- **Classification Metrics:** Each file was assigned an LSB score evaluated against the configured 0.2 threshold. Files exceeding the limit were flagged as suspected steganography carriers.
- **Report Compilation:** The console results were successfully written to report.csv, containing structured columns for filenames, scores, and status flags.

RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

6. RESULT

The steganography detection experiment was successfully executed using StegExpose. The tool scanned the evidence directory via the command line, analyzed combinations of LSB metrics, and generated a clean CSV report detailing the cleanliness or suspicion level of each digital image artifact.